

Cryoscape: AI-Interpreted 3D Ice Printing as Climate Commentary

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Abstract

Cryoscape is an interactive installation that explores the intersection of water's phase transitions, 3D printing technology, and artificial intelligence to create a dynamic commentary on climate change. Born from observations during an Arctic Circle Artist Residency in Svalbard, this project develops a novel approach to ice sculpture through controlled droplet deposition on temperature-regulated surfaces with varying hydrophobic and hydrophilic treatments. The resulting micro-landscapes, captured by real-time macro photography, are interpreted by an AI system generating haiku inspired poetry, creating a dialogue between machine perception and natural phenomena. Each installation adapts to local environmental conditions—humidity, temperature, and water composition—making the work site-specific and collaborative with nature. Beyond its artistic merit, Cryoscape raises critical questions about the anthropogenic impact on Arctic environments, including the ironic contribution of AI systems to global warming through their energy consumption. The technology developed for Cryoscape has potential applications in fields like tissue engineering and microfluidics while serving as a meditation on slow technology, environmental fragility, and the blurred boundaries between artificial and natural systems.

CCS Concepts

• **Applied computing** → Arts and humanities; Media arts; Physical sciences and engineering; Earth and atmospheric sciences; Environmental sciences; • **Computing methodologies** → Artificial intelligence; Natural language processing; Natural language generation.

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Keywords

Ice printing, AI poetry, climate change, interactive installation, slow technology, Arctic art, generative systems, environmental art

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1 Introduction

Water exists in a perpetual cycle of transformation, shifting between vapor, liquid, and solid states in response to environmental conditions. This fundamental property of water serves as both the medium and message in Cryoscape, an interactive installation that employs novel 3D ice printing techniques and artificial intelligence to create ephemeral landscapes that reflect on our changing climate.

The project emerged from direct observations during the Arctic Circle Artist Residency in Svalbard, sailing on a ship among the icy landscape for three weeks, where the omnipresence of water in its various states—vapor in the clouds, snowflakes in the sky, liquid waves in the ocean, the warmth of hot water in our hands, and the solid majesty of glaciers and sea ice. Each phase of water reveals distinct morphologies. We are all connected through bodies of water.

This experience prompted our investigation into whether 3D printing technologies could capture and recreate the intricate morphologies of ice formation while incorporating the temporal dimension of melting as a core aesthetic and conceptual element.

1.1 Context and Motivation

The Arctic region is experiencing warming at twice the global average rate, making it a bellwether for climate change impacts [IPCC, 2021]. The rapid loss of Arctic ice has significant implications for global climate systems, ocean currents, and weather patterns [J and D, 2018]. Artists have long engaged with these environmental



Figure 1: Installation view of Cryoscape at Gallery Ether in Tokyo

concerns, using various media to communicate the urgency and scale of climate change [T.J., 2017].

Recent advances in 3D printing with ice have opened new possibilities for creating temporary structures with complex geometries. Researchers have demonstrated ice printing for applications ranging from artificial blood vessels [J.S et al., 2012] to temporary scaffolds for tissue engineering [M and B, 2015]. However, the artistic potential of these techniques, particularly in creating dynamic, evolving sculptures that respond to environmental conditions, remains largely unexplored.

Simultaneously, the rise of artificial intelligence systems presents both opportunities and contradictions for environmental art. While AI can generate novel interpretations and expand creative possibilities, the energy consumption of training and running large language models contributes significantly to carbon emissions [E et al., 2019]. This paradox—using energy-intensive technology to comment on environmental destruction—forms a critical tension within Cryoscape.

1.2 Related Work

1.2.1 3D Ice Printing and Fabrication. The concept of 3D printing with ice builds upon established additive manufacturing principles while introducing unique challenges related to temperature control and phase transitions [A et al., 2024].

1.2.2 AI-Generated Poetry and Environmental Art.

The intersection of artificial intelligence and poetry has produced compelling works that challenge notions of creativity and

authorship. Projects like "The Seeker" by Allison Parrish [A, 2018] use neural networks to generate poetry from environmental data. Similarly, "AI Haiku" by Ren et al. [X et al., 2019] demonstrates how machine learning can capture the essence of traditional Japanese poetry forms.

In environmental art, data-driven installations have become increasingly sophisticated. Olafur Eliasson's "Ice Watch" [O, 2018] brought glacial ice to urban centers, creating visceral encounters with climate change. Meanwhile, projects like "Pollution Explorers" by Stefanie Posavec [S, 2017] use data visualization to make environmental impacts tangible.

1.2.2 Slow Technology and Contemplative Computing. The concept of "slow technology," introduced by Hallnäs and Redström [L and J, 2001], advocates for designs that encourage reflection rather than efficiency. This philosophy aligns with traditional Japanese aesthetics of *ma* (間) or negative space, and the appreciation of transient beauty in concepts like *mono no aware* (物の哀れ).

Recent works in contemplative computing, such as those by Alex Soojung-Kim Pang [A.S.K, 2013], explore how technology can support rather than disrupt human attention and environmental awareness. Cryoscape embodies these principles through its deliberate pacing—each water droplet taking approximately 20 seconds to freeze—creating a meditative rhythm that contrasts with the typical speed of digital interactions.



Figure 2: View of Cryoscape system setup

1.3 Contribution

Cryoscape advances the field of interactive environmental art through several key contributions:

Novel Ice Printing Technique: We develop a controlled droplet deposition system that creates high-resolution ice sculptures with varying textures based on surface treatments and environmental conditions.

AI-Nature Dialogue: The project establishes a unique feedback loop between natural phenomena and artificial intelligence, where machine-generated haiku poetry interprets evolving ice formations in real-time.

Site-Specific Adaptation: By responding to local environmental conditions (humidity, temperature, water composition), each installation becomes a collaborative work with its location's climate.

Critical Commentary: The work explicitly addresses the paradox of using energy-intensive AI systems to comment on environmental destruction, making visible the hidden costs of digital technologies.

Cross-Disciplinary Applications: Beyond artistic expression, the techniques developed offer potential applications in tissue engineering, microfluidics, and soft robotics.

2 System Design and Implementation

2.1 Hardware Architecture

The Cryoscape system consists of four primary components: the cooling platform, droplet deposition mechanism, imaging, and generative AI system.

2.1.1 Cooling Platform. The base of the system is a thermoelectrically cooled aluminum plate maintained at -15°C ($\pm 0.5^{\circ}\text{C}$). This temperature was selected through extensive experimentation to balance rapid freezing with sufficient time for droplet spreading and crystallization. The plate measures $300\text{mm} \times 300\text{mm}$, providing adequate space for creating varied ice landscapes while maintaining uniform temperature distribution.

Temperature control is achieved through a PID controller connected to an array of Peltier elements beneath the aluminum plate. Heat dissipation occurs through heat sinks.

2.1.2 Droplet Deposition System. Water delivery employs a custom-built five-bar linkage system with a precision syringe pump capable of dispensing droplets ranging from $10\mu\text{L}$ to $500\mu\text{L}$. The

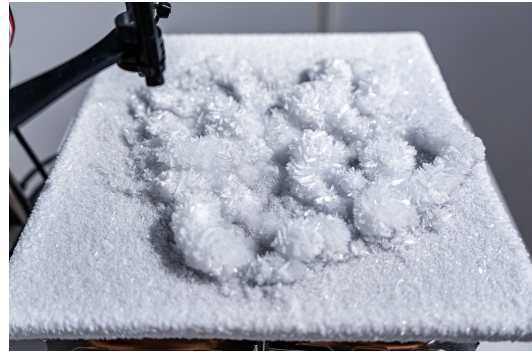


Figure 3: An example of a 3D printed sculpture in-progress

vertical distance between the nozzle and the cooling plate is adjustable from 10mm to 200mm, allowing control over droplet impact velocity and spreading behavior.

The system supports multiple liquid reservoirs, enabling the use of solutions with different freezing points. This capability allows for the creation of temporary support structures using higher melting point liquids, which subsequently melt away as ambient temperature rises.

2.1.3 Surface Treatments. Superhydrophobic coatings create surfaces with contact angles $>150^{\circ}$. These surfaces produce tall, narrow ice structures as droplets maintain spherical shapes during freezing. Hydrophilic surfaces achieve contact angles $<30^{\circ}$, resulting in flat, spreading ice formations.

2.2 Imaging

A motorized macro photography setup captures the evolving ice formations. The camera, mounted on a custom-built five-bar linkage system, performs continuous scanning movements across the cooling plate. The captured imagery feeds to a large screen to display the microscopic ice landscapes in real time, allowing viewers to observe formations invisible to the naked eye. This magnification reveals crystalline structures, dendritic growth patterns, and the interplay of light through translucent ice.

2.3 AI Poetry Generation

The artificial intelligence component employs a fine-tuned language model specifically trained on a corpus of environmental poetry, haiku collections, and scientific descriptions of ice formation. The extracted features from the live feed from the macro camera serve as prompts. The model generates haiku following the traditional 5-7-5 syllable structure while incorporating: seasonal references (kigo) appropriate to ice and water; cutting words (kireji) for emotional resonance; imagery derived from the visual analysis.

The interaction of water with the hydrophobic and hydrophilic treated surfaces produced wildly different structures—some resembling lunar landscapes, others evoking the snowy expanses of Hokkaido, the crystalline intricacies of microscopic images, or even the forms found in Noguchi Playgrounds or the paintings of Kåre Tveter.

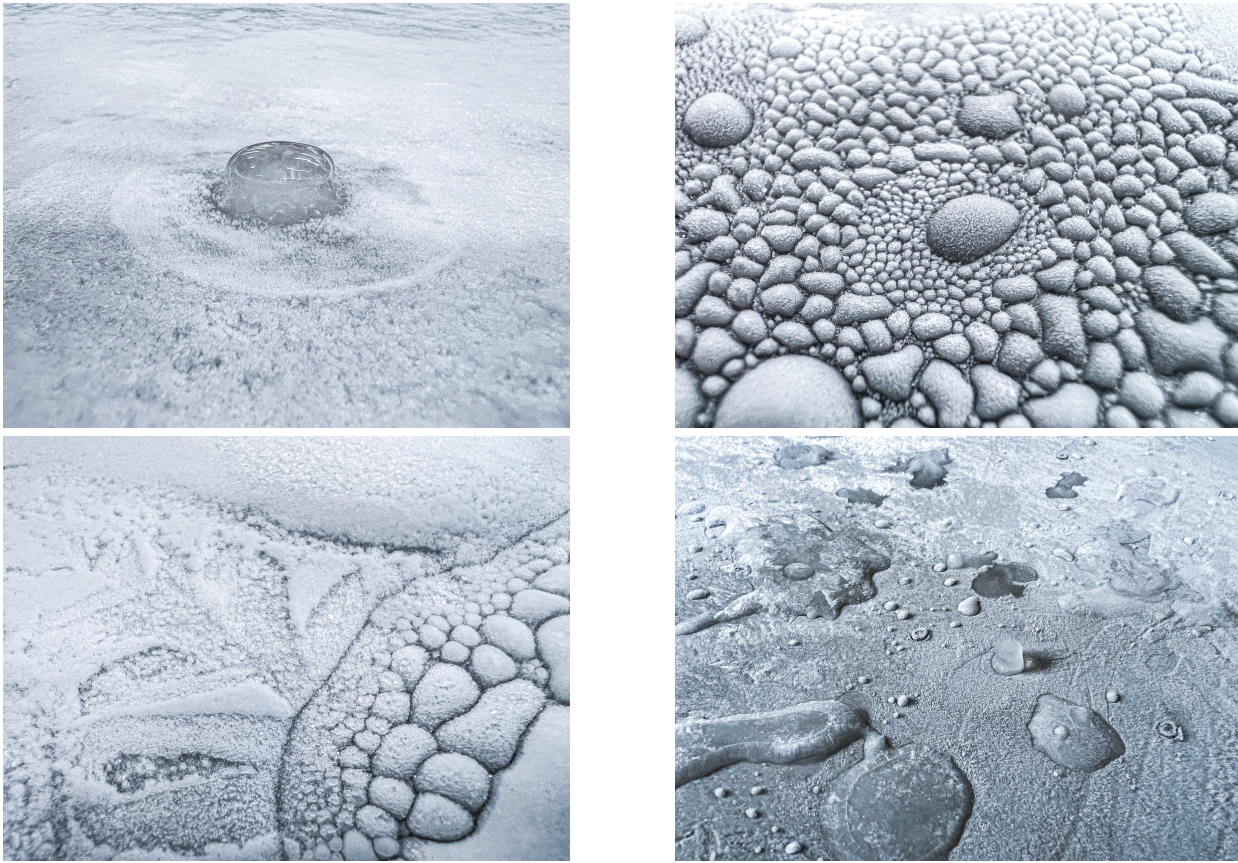


Figure 4: Example images captured from the macro camera feed that the AI system use to write Haiku

Figure 4 presents examples of haiku generated from the evolving ice textures captured by the macro camera. Each poem reflects the visual qualities of the corresponding micro-landscape.

Top left:

Alien relic's glow,
Timeless echo in starlight,
Silent worlds below.

Top right (Surface treated with hydrophobic spray):

Atoms dream of storms,
Stones pulse with celestial breath,
Worlds within worlds spin.

Bottom left (Surface treated with hydrophobic spray on the right):

Ancient stream whispers,
Winding through a dreamscape's veil,
Pebbles' secret dance.

Bottom right:

Silent lunar plain,
Echoes of celestial tears,
Dreams on ice remain.

2.4 Interactive Control Panel

We implemented an interface for controlling the path of droplet deposition. In the Processing sketch, users can choose between

random deposition or the recording and playback of a specific path—such as the contour of an Arctic island, daily ice chart of Svalbard, or the crystalline profile of a snowflake. The selected path's XY coordinates are stored and translated into servo motor angles using inverse kinematics. These angles are then transmitted to a microcontroller, which actuates a five-bar linkage mechanism spanning the cooling platform. Equipped with a dripping nozzle and a microscopic camera, the system effectively transforms into 3D ice printing, enabling live visual feedback of the deposition process.

3 Artistic Concept and Climate Reflection

3.1 Water as Universal Medium

Water serves as the fundamental connector across scales—from molecular hydrogen bonds to global ocean currents. In Cryoscape, this universality manifests through the transformation of individual droplets into collective landscapes, mirroring how local climate actions aggregate into global effects. The choice to work with water's phase transitions directly references the critical role of ice in Earth's climate system. The albedo effect of polar ice, thermohaline circulation driven by freezing and melting, and the release of trapped



Figure 5: Installation view of Cryoscape



Figure 6: Detailed view of ice crystallization on Cryoscape

greenhouse gases from permafrost all find abstract representation in the micro-scale processes visible in the installation.

3.2 Temporal Dimensions

Cryoscape operates across multiple timescales:

- Immediate: The 20-second freezing cycle of individual droplets
- Experiential: The minutes or hours viewers spend observing formations
- Installation: Days or weeks of continuous operation at each site
- Geological: Reference to millennia-old ice cores and glacial cycles

This temporal layering encourages viewers to consider their position within larger environmental cycles, moving beyond human-centric timescales to contemplate geological and climatic temporalities.

3.3 The Melting as the Message

Cryoscape's core concept is the inevitability of melting. Unlike traditional sculpture that aspires to permanence, these ice formations exist in constant flux. The melting process becomes a core aesthetic element, with different regions of the sculpture disappearing at varying rates based on their composition and structure. This ephemeral quality serves multiple metaphorical functions: it offers commentary on the impermanence of Arctic ice, reflects on the fragility of ecological systems, meditates on the concepts of

impermanence (*mujo*), and critiques art market preferences for permanent, collectable objects.

3.4 AI as Environmental Paradox

The integration of artificial intelligence introduces a critical tension within the work. Training large language models requires enormous computational resources, with some models consuming as much energy as hundreds of homes use annually [D et al., 2021]. The carbon footprint of AI development thus contributes directly to the climate change the project critiques.

We made this paradox explicit by showing the real-time power consumption of the AI system, calculated carbon emissions, and comparisons to the melting rate of Arctic ice. Viewers confront the reality that their aesthetic experience contributes incrementally to the environmental crisis being portrayed.

The AI's role as interpreter of natural phenomena also raises questions about technological mediation of environmental experience. As direct encounters with wilderness become rarer, are we increasingly reliant on technological interfaces to understand nature? The haiku generated by observing ice formations suggests both the potential for technology to inspire environmental awareness and the risk of further separation from direct natural experience.

This paradox becomes particularly acute when considering Alaska's emerging role in AI infrastructure. The state is actively discussing building data centers for AI in the North Slope, attracted by the cool climate that reduces cooling costs, vast available land, and access to oil-based energy resources [Beacon, 2024]. The massive energy requirements of these proposed AI data centers would transform Alaska—home to rapidly melting glaciers—into a contributor to the very computational systems accelerating climate change. This ironic feedback loop exemplifies the contradictions embedded in technological solutions to environmental crises.

3.5 Site-Specificity and Nature Co-creation

Cryoscape evolves as it travels to different cities. In Alaska, Tokyo, Austin, Vancouver, Linz, and beyond, the humidity, temperature, air quality, and water calcium unique to each city shape the ice sculptures, making each iteration site-specific and collaborative with nature. The installation becomes a form of climate portraiture, revealing invisible atmospheric qualities through their effects on ice formation.

3.6 Who is the Artist?

This project raises fundamental philosophical questions about authorship and authenticity. Who is the artist here? Is it the AI system generating poetry, the environmental data shaping formations, the moisture and minerals of each city, or the human creators who assembled these elements? The distributed agency challenges traditional notions of artistic authorship, suggesting instead a network of human, non-human, and more-than-human collaborators.

4 Discussion

Cryoscape is an artistic inquiry into climate change, employing technical innovation as a means rather than an end.

4.1 Nature in Technological Contexts

What constitutes "nature" when ice formations are created in a lab-controlled environment? The AI searches for nature-likeness in formations that are simultaneously natural (following physical laws of crystallization) and artificial (created through technological intervention). By using an extremely human-made technology—AI—to interpret and define nature, this project challenges the boundaries between artificial and natural, human and non-human.

The boundaries between natural and artificial are getting increasingly meaningless in the Anthropocene, where human influence extends to all Earth systems. Rather than maintaining false dichotomies, Cryoscape suggests a hybrid aesthetic acknowledging technology's inextricable entanglement with environmental processes.

4.2 Slow Technology

In an era of accelerating technological change and instantaneous digital gratification, Cryoscape's deliberate pace offers a form of temporal resistance. The 20-second freezing cycle cannot be rushed; viewers must synchronize with the material's inherent temporality. This enforced slowness serves multiple functions: it encourages sustained attention and contemplation, mirrors the slow violence of climate change, and emphasizes process over product. The work demonstrates that technological art need not perpetuate cultures of speed and efficiency, instead offering alternative relationships with time and attention.

4.3 Ethics of Energy Use in Climate Art

The paradox of using energy-intensive systems to critique climate change raises ethical questions extending beyond this project. How do we balance the carbon footprint of creating and exhibiting environmental art against its potential for awareness-raising and behavior change? Cryoscape addresses this dilemma through transparency about its own energy consumption and explicit thematization of the paradox within the work. Rather than claiming carbon neutrality or ignoring these contradictions, the project foregrounds them as part of its critical framework.

5 Conclusion

Cryoscape generates productive tensions between permanence and transience, natural and artificial, local and global, human and machine perception—tensions that reflect the complexities of addressing climate change in an era of technological acceleration. As the project continues to evolve across different sites and seasons, it serves as both a portrait of our changing climate and a meditation on technology's role in environmental futures, demonstrating that innovative artistic applications of emerging technologies can create meaningful dialogue about our planetary crisis while acknowledging their own complicity in these systems. The techniques developed for Cryoscape extend beyond artistic exploration, offering potential applications in fields like tissue engineering, microfluidics, and soft robotics, where ice as a biocompatible material can serve as a temporary support structure for 3D printing, easily melting away after use to enable precise and adaptable designs. Ultimately, Cryoscape asks viewers to consider their own position within cycles of freezing and melting, creation and destruction—in watching

water transform from liquid to solid and back again, we witness a fundamental process that connects molecular interactions to planetary systems, a reminder that we are all, indeed, bodies of water in constant flux.

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